MAT 3007 Homework # 5 Spring 2024

Due: Wednesday, April 10 at 12:00pm (noon)

Please note: You should submit your answers (scanned or typeset) in a PDF file. Question 2 and 3 asks you to write codes in Python or MATLAB. Submit all necessary code files via Blackboard. Also, give meaningful names to your decision variables and constraints, and add comments to your code liberally. The problem set is due at 12:00pm (noon) on Wednesday, April 10th.

Hint: In MATLAB you can use the following commands to plot the surface/contour figure:

```
[X1,X2] = meshgrid(-1:.1:1, -1:.1:1);
F = X1 .* X2;
surf(X1,X2,F);
contour(X1,X2,F);
```

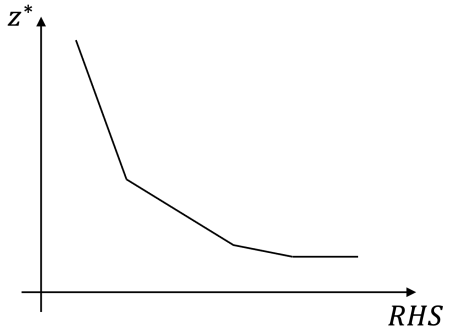
In Python you can use the following commands to plot the surface/contour figure:

```
import numpy as np
import matplotlib.pyplot as plt
X1 = np.arange(-1, 1, 0.2)
X2 = np.arange(-1, 1, 0.2)
X1, X2 = np.meshgrid(X1, X2)
F = X1 * X2

# plot the surface figure
fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')
ax.plot_surface(X1, X2, F)

# plot the contour figure
fig, ax = plt.subplots()
ax.contour(X1, X2, F)
```

1. (20 pts) The following graph depicts the optimal value of a linear program, denoted by z^*, as the right-hand side of one of the constraints, denoted by RHS, changes.



(a) Determine whether this LP is a maximization or minimization problem, and whether the constraintis a "$\geq$" or a "$\leq$" constraint. Explain your reasons.

(b) Now replace "RHS" by "objective coefficient" and interpret this graph depicting how the optimal objective changes as the objective coefficient of a primal variable is changed. Determine if the linear program is a maximization or minimization problem, and whether the (optimal) value of the primal variable is nonnegative or nonpositive.

2. (30 pts) Consider the unconstrained optimization problem

$$\min_{x\in\mathbb{R}^2} \quad f(x) \coloneqq x_1^4 + \frac{2}{3}x_1^3 + \frac{1}{2}x_1^2 - 2x_1^2x_2 + \frac{4}{3}x_2^2.$$

(a) Calculate all stationary points of f and investigate whether the stationary points are local maximizer, local minimizer, or saddle points.

(b) Create a surface and/or contour plot of the function using MATLAB or Python and decide whether the problem possesses a global solution or not.

3. (30 pts) For the following function

$$f(x) = x_1 e^{-\frac{1}{2}(x_1^2+x_2^2)}.$$

(a) Write down the gradient and the Hessian matrix of $f(x)$.

(b) Derive the second-order Taylor expansion of $f(x)$ at the point $\bar{x} = [-1, \frac{1}{2}]$.

(c) Plot the function in part (a) in the area $x_1 \in [-3, 3]$ and $x_2 \in [-3, 3]$

(d) Plot the second-order Taylor expansion in part (b) in the same figure in the area $x_1 \in [-1.5, -0.5]$ and $x_2 \in [-1, 1]$. The goal is to demonstrate that the Taylor expansion approximates $f(x)$ very well close to $\bar{x}$, so please print the figure from an angle that helps to see how the function and the Taylor expansions compare.

4. (20 pts) Consider the nonlinear optimization problem:

$$\begin{aligned} \min_{x_1,x_2,x_3} \quad & 2x_1x_2 + \frac{1}{2}x_3^2 \\ \text{s.t.} \quad & 2x_1x_3 + \frac{1}{2}x_2^2 \leq 0 \\ & 2x_2x_3 + \frac{1}{2}x_1^2 \leq 0. \end{aligned}$$

(a) Write down the KKT conditions for this problem.

(b) Investigate whether the point $x^* = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ is a KKT point.